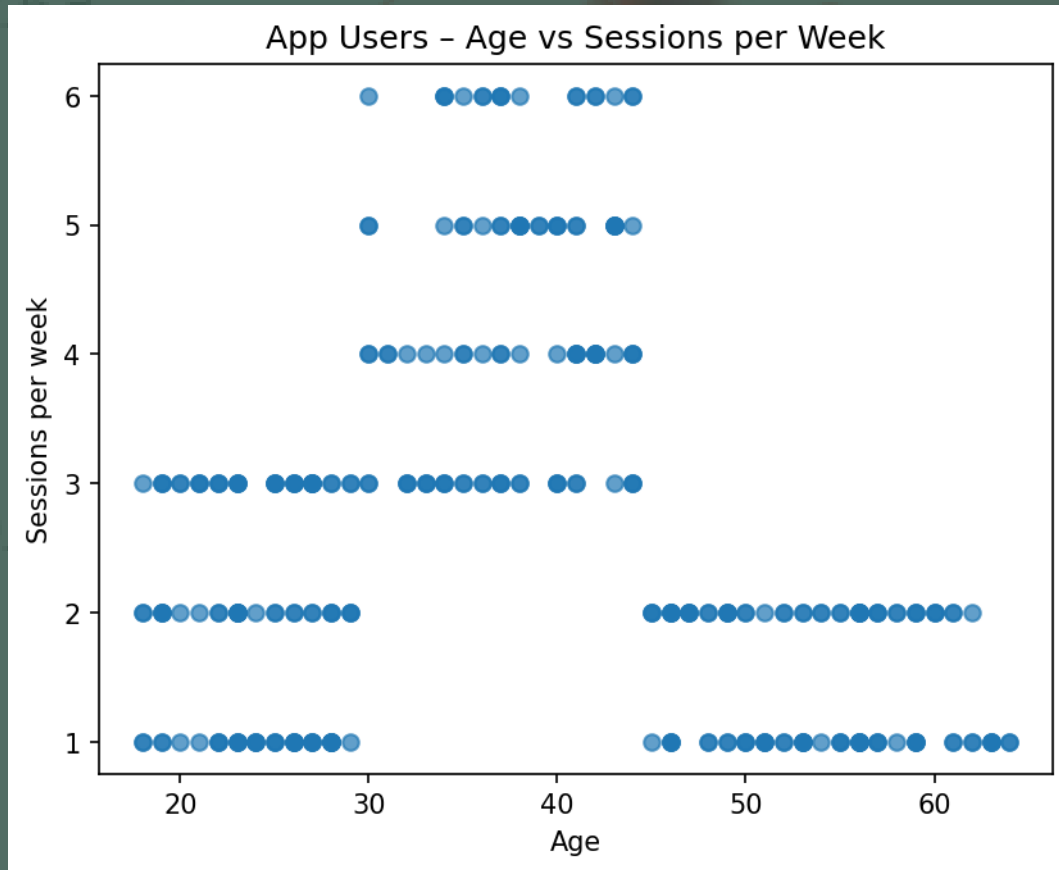


CUSTOMER SEGMENTATION

Kmeans Clustering for Customer Segmentation

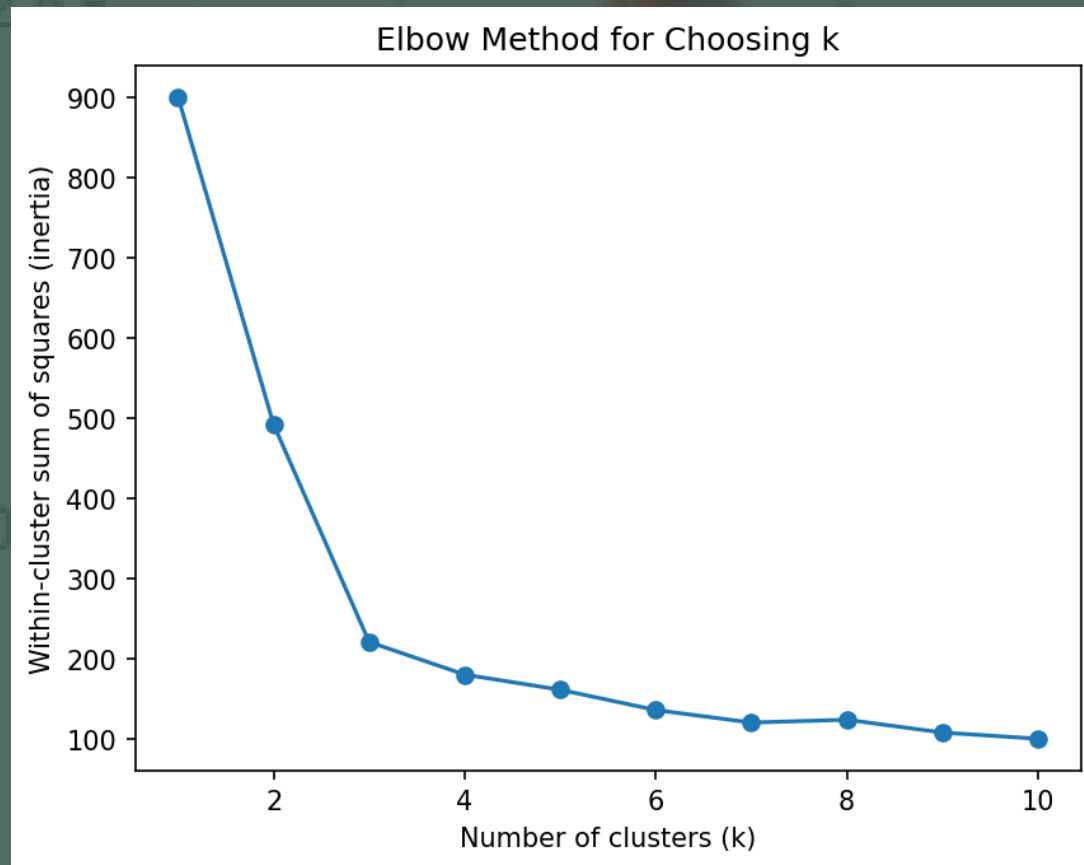


Why We Need Behaviour-Based Segmentation



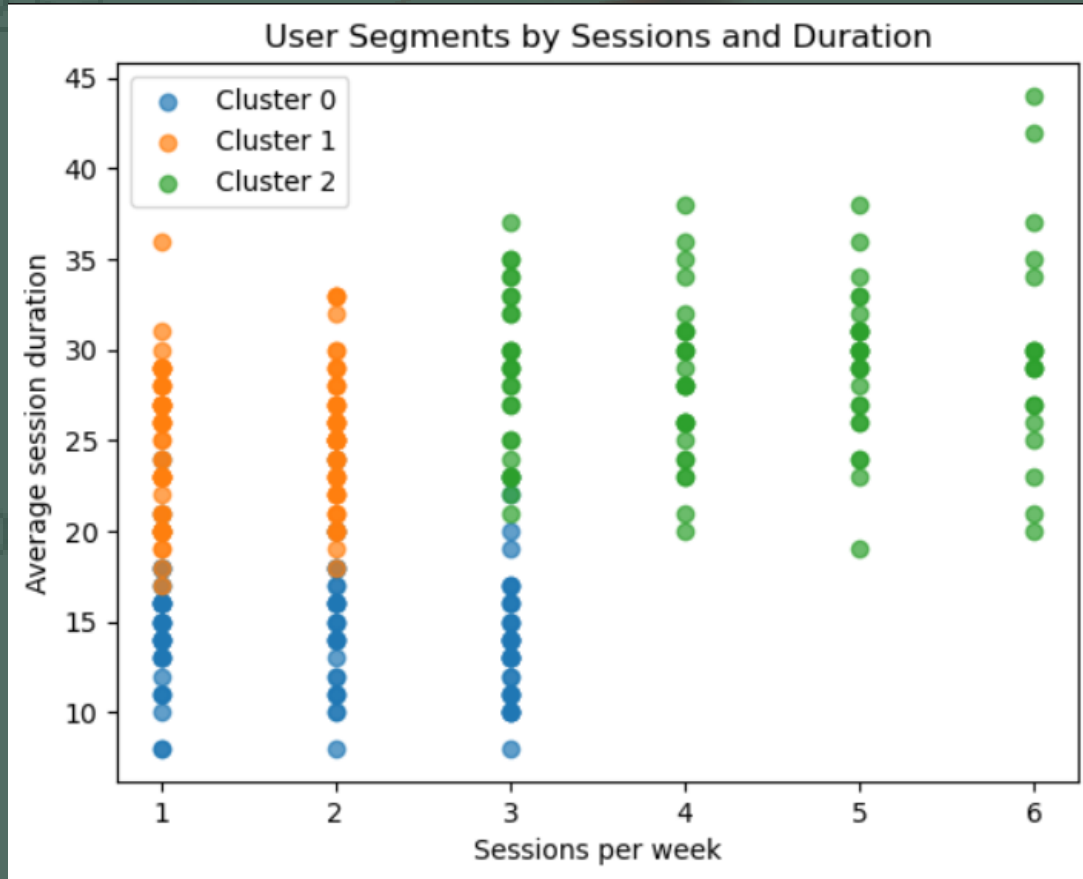
- One app, many very different usage patterns
- Users vary by age, visit frequency and session length
- “One-size-fits-all” engagement is leaving value on the table

Choosing the Right Number of Segments



- Tested 1–10 clusters using K-Means
- Elbow point around $k \approx 3$
- 3 segments balance simplicity and insight

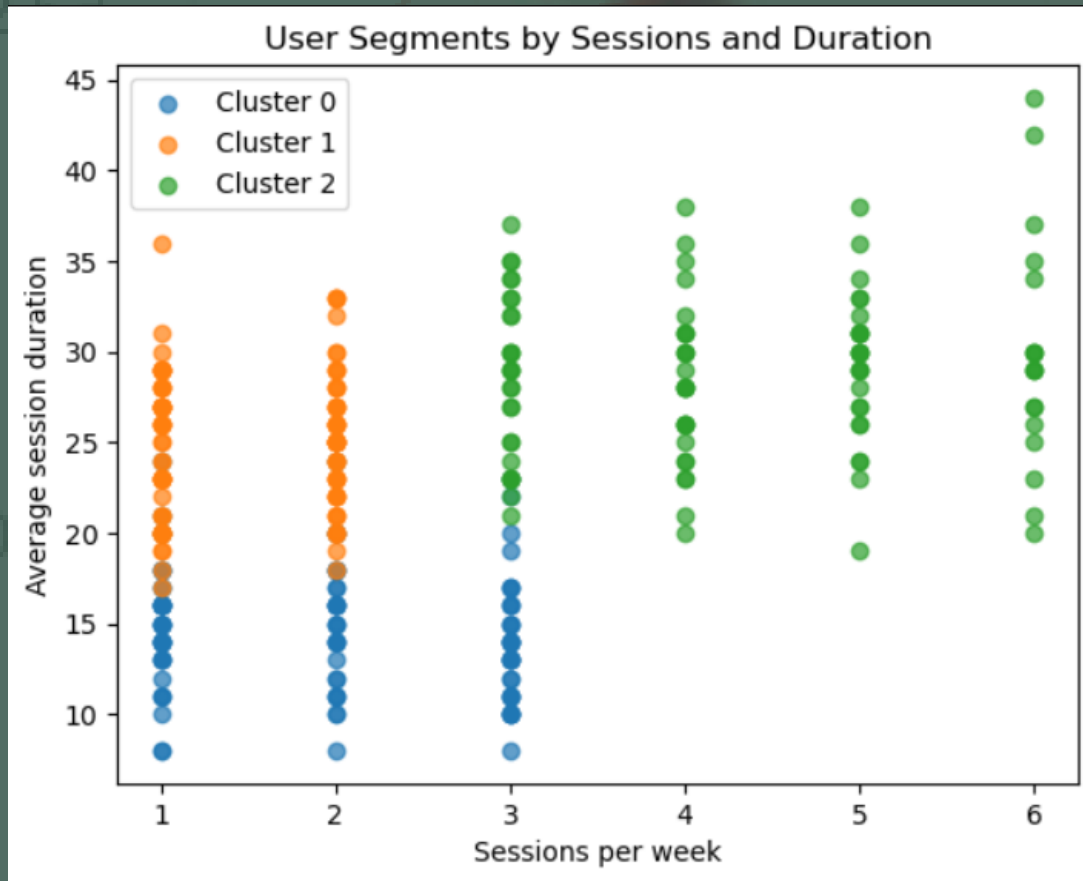
Three Distinct User Behaviours



Sessions per week vs Avg session duration, coloured by cluster

- Cluster 0 – Light, quick users
- Cluster 1 – Occasional but deep users
- Cluster 2 – Heavy, highly engaged users

What Makes Each Segment Different?



Bar chart of average age, sessions_per_week, avg_session_duration by cluster

- Cluster 0: low frequency, short sessions → nurture
- Cluster 1: low–mid frequency, long sessions → deepen
- Cluster 2: high frequency, long sessions → retain & leverage

Business Impact and Next Steps

- Design 3 tailored engagement strategies (one per segment)
- Run A/B tests on messaging and offers by segment
- Track uplift: retention, sessions per week, and revenue per user

